

EASTERN INTERNATIONAL UNIVERSITY

SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

DEPARTMENT OF COMPUTER NETWORKS AND DATA COMMUNICATIONS



PROJECT REPORT

Design a VPN (MPLS) Network

Protocols: MPLS, LDP, BGP, OSPF, VRF

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Binh Duong, December 2025

Abstract

This project explores the design and implementation of a Virtual Private Network (VPN) using Multiprotocol Label Switching (MPLS) technology. As modern organizations require secure, reliable, and scalable communication between remote sites, traditional VPN solutions often face limitations in performance and scalability. This report presents the simulation of an enterprise MPLS VPN topology using PNETLab software. The project involves configuring essential protocols such as LDP, BGP, and OSPF to establish label distribution and routing. Furthermore, Layer 3 VPN (L3VPN) services are implemented to interconnect branches, and VRF technology is used for traffic separation. The results verify network connectivity, Quality of Service (QoS), and the efficiency of data transfer across the simulated environment.

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List of Abbreviations

Abbreviation	Full word
BGP	Border Gateway Protocol
CE	Customer Edge (Router)
FIB	Forwarding Information Base
IT	Information Technology
L3VPN	Layer 3 Virtual Private Network
LDP	Label Distribution Protocol
LER	Label Edge Router
LFIB	Label Forwarding Information Base
LSR	Label Switching Router
MP-BGP	Multiprotocol Border Gateway Protocol
MPLS	Multiprotocol Label Switching
OSPF	Open Shortest Path First
P	Provider (Router)
PE	Provider Edge (Router)
QoS	Quality of Service
RD	Route Distinguisher
RT	Route Target
VPN	Virtual Private Network
VRF	Virtual Routing and Forwarding

CHAPTER 1

Introduction

1.1 Motivation

In the modern era of digital transformation, enterprise networks serve as the backbone for critical business operations, requiring secure, reliable, and scalable communication channels between headquarters and geographically distributed branches. As organizations expand, the demand for high-performance connectivity increases, necessitating robust network architectures that can handle diverse traffic types with varying Quality of Service (QoS) requirements.

Traditional Virtual Private Network (VPN) solutions often face significant limitations in scalability, performance efficiency, and management complexity. As the network grows, maintaining a mesh of traditional VPN tunnels becomes administratively burdensome and less efficient in traffic engineering. Consequently, Multiprotocol Label Switching (MPLS) has emerged as a standard technology for service provider networks, offering a unified approach to transporting multiple traffic types.

Designing a VPN based on MPLS technology provides a superior solution by combining the scalability of IP routing with the performance of Layer 2 switching. This approach not only improves traffic management and enhances security but also ensures strict traffic separation between different departments or customers using Virtual Routing and Forwarding (VRF).

Motivated by the practical importance of this technology in the telecommunications industry, this project focuses on simulating an enterprise-grade MPLS VPN environment. By implementing this topology, the study develops both a technical understanding of label switching mechanisms and practical skills in configuring complex routing protocols.

1.2 Project Objectives

The primary objective of this project is to design, simulate, and analyze a fully functional MPLS VPN network that ensures secure and efficient data transfer between remote branches.

The specific objectives are as follows:

- **Design and simulation:** design a comprehensive enterprise network topology and simulate it using PNETLab software with Cisco router images.
- **Protocol implementation:** configure the core MPLS backbone using Label Distribution Protocol (LDP) for label exchange and Open Shortest Path First (OSPF) as the Interior Gateway Protocol (IGP).
- **VPN configuration:** implement Layer 3 VPN (L3VPN) services to interconnect branch offices, using Multiprotocol BGP (MP-BGP) for distributing VPNv4 routes and VRF for traffic isolation.
- **Verification and analysis:** verify network connectivity, ensure proper traffic separation between different VRF instances, and analyze forwarding paths using simulation tools.

1.3 Organization of the Report

The report is organized into five chapters:

- **Chapter 1** introduces the motivation, problem context, and objectives of the project.
- **Chapter 2** presents the theoretical background related to MPLS, LDP, VRF, and MP-BGP.
- **Chapter 3** describes the simulation environment, network design, addressing plan, and device configuration.
- **Chapter 4** reports the simulation results and analyzes operational verification outputs.
- **Chapter 5** summarizes the findings and suggests future directions for development.

CHAPTER 2

Theoretical Background

2.1 Introduction to MPLS Technology

Multiprotocol Label Switching (MPLS) is a protocol-agnostic routing technique designed to speed up and shape traffic flows across enterprise wide-area networks and service provider infrastructures. Unlike traditional IP routing, which requires each router to perform a route lookup based on the destination IP address, MPLS forwards data using short path labels.

In an MPLS network, the ingress router encapsulates the IP packet with an MPLS header. Subsequent Label Switching Routers (LSRs) read this label to make forwarding decisions without inspecting the original IP header. This mechanism improves forwarding speed and supports advanced traffic engineering.

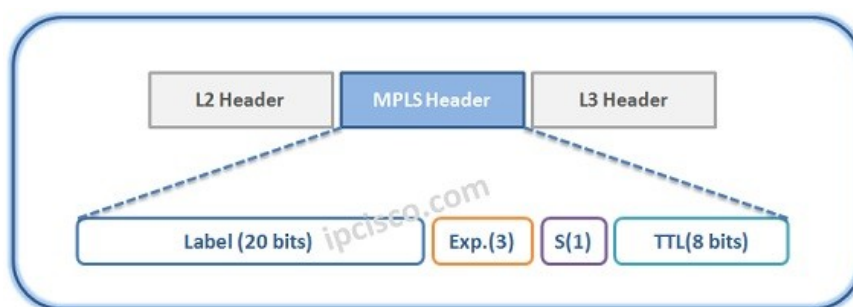


Figure 2.1: Standard MPLS Header Structure

The MPLS header is 32 bits long and is inserted between Layer 2 and Layer 3 headers. It consists of four fields:

- **Label (20 bits):** the actual label value used for forwarding.
- **Exp (3 bits):** experimental bits commonly used for QoS.
- **S (1 bit):** bottom-of-stack bit indicating the last label in the stack.
- **TTL (8 bits):** time to live, used to prevent looping.

2.2 Label Distribution Protocol (LDP) Operations

For MPLS to function, routers must agree on the meaning of labels. The Label Distribution Protocol (LDP) is the standard IETF signaling protocol used to distribute label bindings between MPLS-enabled routers.

LDP operates by establishing sessions between neighboring routers. Once a session is established, routers exchange label mappings for the networks they know.

- **Hello messages:** used to discover neighbors and typically sent to multicast address 224.0.0.2.
- **Session establishment:** uses TCP port 646 for reliable communication.

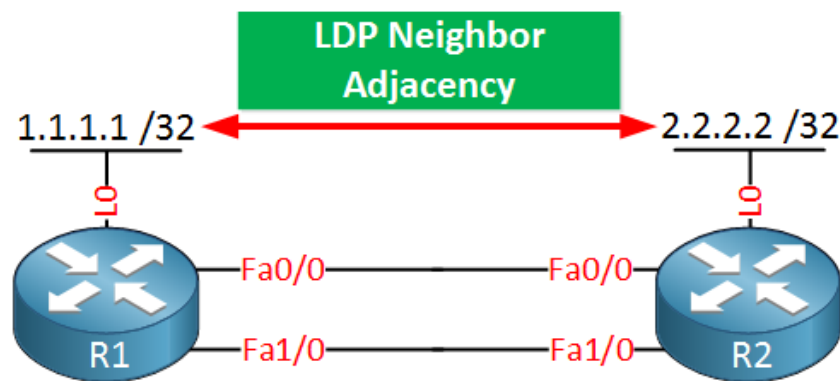


Figure 2.2: LDP Neighbor Discovery and Session Establishment

2.3 MPLS Layer 3 VPN Architecture

MPLS Layer 3 VPN (L3VPN) is among the most common applications of MPLS technology. It allows a service provider to offer IP VPN services to customers over a shared backbone while keeping customer traffic isolated.

The architecture consists of three main device roles:

- **Customer Edge (CE):** a customer-side router that connects to the service provider and uses standard IP routing.
- **Provider Edge (PE):** an edge router that connects to CE routers and maintains VRF tables while running MP-BGP.
- **Provider (P):** a core router that performs label switching and does not carry customer routes.

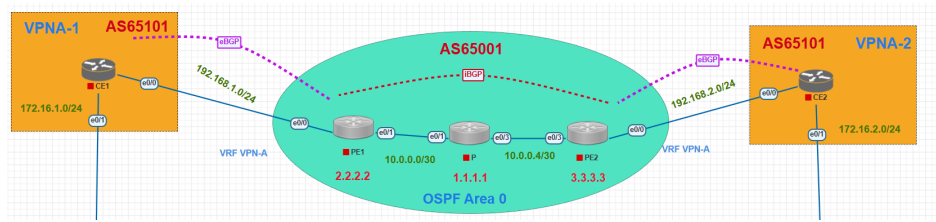


Figure 2.3: MPLS L3VPN Network Architecture

2.4 Virtual Routing and Forwarding (VRF)

Virtual Routing and Forwarding (VRF) allows multiple routing-table instances to coexist on the same router. Since each routing instance is independent, the same or overlapping IP addresses may be used without conflict.

In an MPLS VPN environment, each customer connects to a specific interface on the PE router. That interface is assigned to a unique VRF, ensuring that traffic from one customer is looked up only in its corresponding routing table and remains isolated from other customers.

2.5 Multiprotocol BGP (MP-BGP) for VPNv4

Standard BGP carries only IPv4 routes. In an MPLS VPN, however, different customers may use overlapping private IPv4 address spaces. To solve this ambiguity, MPLS VPN uses Multiprotocol BGP (MP-BGP).

MP-BGP adds a unique Route Distinguisher (RD) to the customer IPv4 prefix, creating a globally unique 96-bit VPNv4 address.

- **Route Distinguisher (RD):** a 64-bit value prepended to the IPv4 address, for example '65001:10'.
- **Route Target (RT):** an extended community attribute controlling route import and export policy for a VRF.

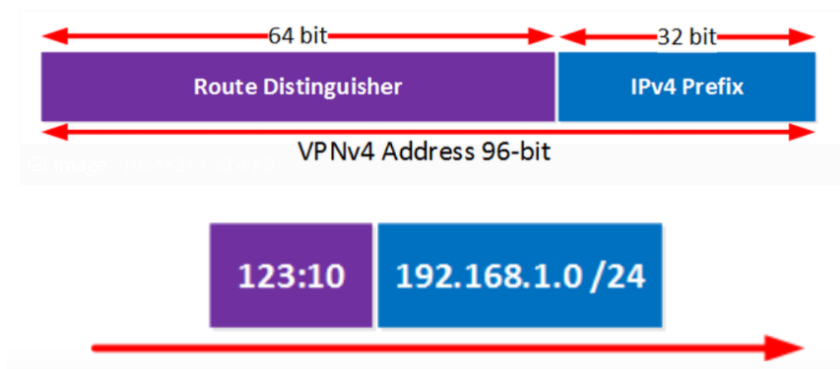


Figure 2.4: VPNv4 Address Structure using Route Distinguisher

CHAPTER 3

System Design and Implementation

3.1 Simulation Environment

To validate the theoretical concepts of MPLS VPN and ensure the feasibility of the design before actual deployment, this project adopts a network emulation approach. This method allows the integration of real network operating systems and therefore provides a higher degree of fidelity than conventional packet simulators.

3.1.1 Introduction to PNETLab

PNETLab (Packet Network Emulator Tool) is a platform that allows network engineers to design and simulate complex network topologies. Unlike basic simulators that only mimic device behavior, PNETLab uses hypervisor technology to run actual router images.

For this project, PNETLab is selected because it:

- supports Cisco IOS images needed for MPLS-related configuration;
- provides a visual workspace for device interconnection and link monitoring; and
- supports traffic capture, including Wireshark integration, for packet-level analysis.

3.1.2 Device Selection

The topology is built using simulated Cisco devices as follows:

- **Provider and Provider Edge routers:** simulated Cisco IOSv Layer 3 routers that support MPLS, LDP, and MP-BGP.
- **Customer Edge routers:** Cisco IOSv routers acting as branch gateways and peering with the provider via eBGP.
- **Switching infrastructure:** Cisco IOSvL2 switches connecting end-user devices within each branch LAN.

- **End devices:** VPCS hosts used for connectivity testing through ping and traceroute.

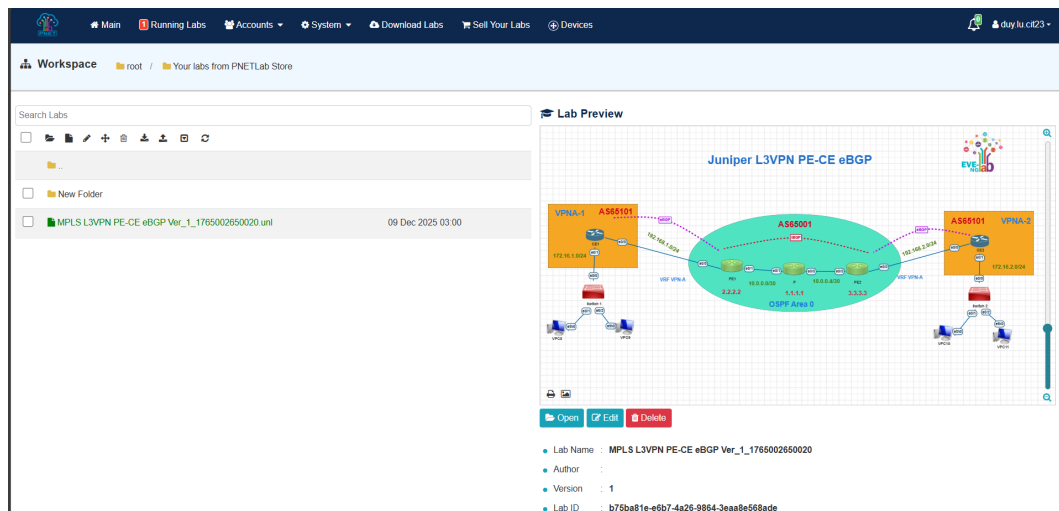


Figure 3.1: PNETLab Workspace Interface

3.2 Network Topology Design

3.2.1 Topology Overview

The designed network topology replicates a typical service-provider architecture connecting two remote customer branches under a single VPN instance (VPN-A).

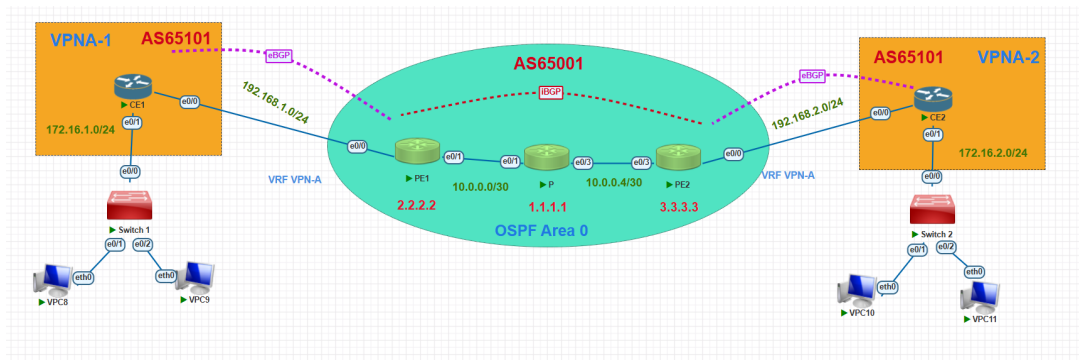


Figure 3.2: MPLS L3VPN Network Topology

The topology includes the following components:

- **Service Provider Core (AS 65001):** the P router (1.1.1.1) performs label switching, while PE1 (2.2.2.2) and PE2 (3.3.3.3) serve as ingress and egress edge routers.
- **Customer Sites (AS 65101):** CE1 connects Branch 1 (172.16.1.0/24) to PE1, while CE2 connects Branch 2 (172.16.2.0/24) to PE2.

3.2.2 IP Addressing Plan

Table 3.1: IP Addressing Scheme

Device	IPv4	Subnet Mask	Description
P	E0/1: 10.0.0.2	255.255.255.252	Link to PE1
P	E0/3: 10.0.0.5	255.255.255.252	Link to PE2
P	Loopback0: 1.1.1.1	255.255.255.255	Router ID / MPLS ID
PE1	E0/0: 192.168.1.2	255.255.255.0	Link to CE1
PE1	E0/1: 10.0.0.1	255.255.255.252	Link to P
PE1	Loopback0: 2.2.2.2	255.255.255.255	Router ID / BGP peer
PE2	E0/0: 192.168.2.2	255.255.255.0	Link to CE2
PE2	E0/3: 10.0.0.6	255.255.255.252	Link to P
PE2	Loopback0: 3.3.3.3	255.255.255.255	Router ID / BGP peer
CE1	E0/0: 192.168.1.1	255.255.255.0	Link to PE1
CE1	E0/1: 172.16.1.1	255.255.255.0	LAN gateway for Site 1
CE2	E0/0: 192.168.2.1	255.255.255.0	Link to PE2
CE2	E0/1: 172.16.2.1	255.255.255.0	LAN gateway for Site 2
VPC 8	172.16.1.10	255.255.255.0	Host at Site 1
VPC 9	172.16.1.11	255.255.255.0	Host at Site 1
VPC 10	172.16.2.10	255.255.255.0	Host at Site 2
VPC 11	172.16.2.11	255.255.255.0	Host at Site 2

3.3 MPLS Backbone Configuration

The first phase involves setting up the service-provider backbone. All core routers (PE1, P, and PE2) must achieve full IP reachability before MPLS is enabled.

3.3.1 IGP Configuration (OSPF Area 0)

OSPF Area 0 is used as the Interior Gateway Protocol to advertise loopback addresses and physical links within the core.

Configuration on P Router

```
P(config)# router ospf 1
P(config)# router-id 1.1.1.1
P(config-router)# network 1.1.1.1 0.0.0.0 area 0
P(config-router)# network 10.0.0.0 0.0.0.3 area 0
P(config-router)# network 10.0.0.4 0.0.0.3 area 0
```

Configuration on PE1 Router

```
PE1(config)# router ospf 1
PE1(config)# router-id 2.2.2.2
PE1(config-router)# network 2.2.2.2 0.0.0.0 area 0
```



```
PE1(config-router)# network 10.0.0.0 0.0.0.3 area 0
```

Configuration on PE2 Router

```
PE2(config)# router ospf 1
PE2(config)# router-id 3.3.3.3
PE2(config-router)# network 3.3.3.3 0.0.0.0 area 0
PE2(config-router)# network 10.0.0.4 0.0.0.3 area 0
```

3.3.2 Enabling MPLS and LDP on Core Interfaces

MPLS is enabled on interfaces participating in the core network ('PE1-P' and 'P-PE2'). LDP is then used to distribute labels for loopback reachability.

Configuration on PE1 (core-facing interface)

```
PE1(config)# interface e0/1
PE1(config-if)# description Link-to-P
PE1(config-if)# ip address 10.0.0.1 255.255.255.252
PE1(config-if)# mpls ip
```

Note: MPLS is not configured on customer-facing interfaces.

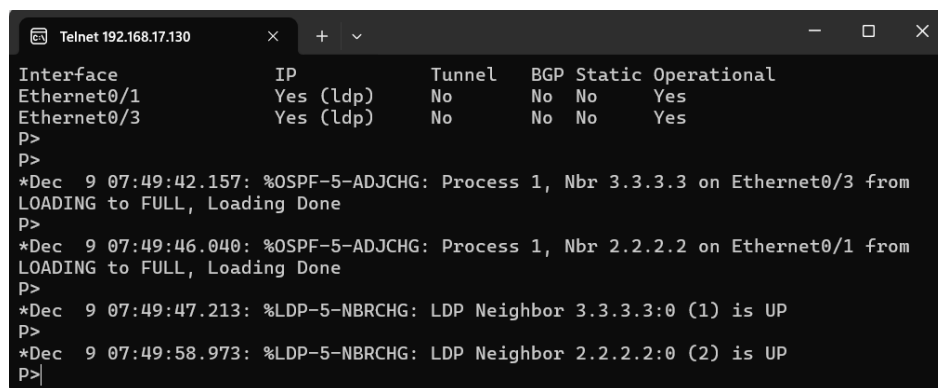


Figure 3.3: Verifying MPLS Enabled Interfaces on the P Router

3.4 Layer 3 VPN Configuration

This phase establishes the private tunnel for customer traffic over the MPLS backbone.

3.4.1 VRF Definition and Route Distinguishers

VRF instances are created on PE routers to isolate customer traffic.

- **Route Distinguisher (RD):** 65001:1 identifies the VPN instance.

- **Route Target (RT):** 65001:1 defines route import and export policy.

Configuration on PE1

```
PE1(config)# ip vrf VPNA
PE1(config-vrf)# rd 65001:1
PE1(config-vrf)# route-target export 65001:1
PE1(config-vrf)# route-target import 65001:1

PE1(config)# interface e0/0
PE1(config-if)# description Link-to-CE1
PE1(config-if)# ip vrf forwarding VPNA
PE1(config-if)# ip address 192.168.1.2 255.255.255.0
```

3.4.2 MP-BGP Configuration Between PE Routers

MP-BGP is configured between PE1 and PE2 to exchange VPNv4 routes through an iBGP session established over loopback addresses.

```
PE1(config)# router bgp 65001
PE1(config-router)# bgp router-id 2.2.2.2
PE1(config-router)# no bgp default ipv4-unicast
PE1(config-router)# neighbor 3.3.3.3 remote-as 65001
PE1(config-router)# neighbor 3.3.3.3 update-source Loopback0
PE1(config-router)# address-family vpnv4
PE1(config-router-af)# neighbor 3.3.3.3 activate
PE1(config-router-af)# neighbor 3.3.3.3 send-community extended
PE1(config-router-af)# exit-address-family
```

```
0:PE1
PE1>
PE1>show bgp vpnv4 unicast all summary
BGP router identifier 2.2.2.2, local AS number 65001
BGP table version is 4, main routing table version 4
2 network entries using 312 bytes of memory
2 path entries using 168 bytes of memory
3/2 BGP path/bestpath attribute entries using 504 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
1 BGP extended community entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1032 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ  OutQ Up/Down  State/PfxRcd
3.3.3.3        4      65001    34     34      4     0    0 00:27:51      1
192.168.1.1    4      65101    35     34      4     0    0 00:28:38      1
```

Figure 3.4: MP-BGP Session Establishment between PE Routers

3.4.3 PE-CE Routing Configuration

Routing is configured between Provider Edge and Customer Edge devices to exchange LAN prefixes. In this design, eBGP is used between each CE and PE pair.

Configuration on PE1 (VRF address family)

```
PE1(config)# router bgp 65001
PE1(config-router)# address-family ipv4 vrf VPNA
PE1(config-router-af)# neighbor 192.168.1.1 remote-as 65101
PE1(config-router-af)# neighbor 192.168.1.1 activate
PE1(config-router-af)# neighbor 192.168.1.1 as-override
PE1(config-router-af)# exit-address-family
```

Configuration on CE1

```
CE1(config)# router bgp 65101
CE1(config-router)# neighbor 192.168.1.2 remote-as 65001
CE1(config-router)# network 172.16.1.0 mask 255.255.255.0
```

CHAPTER 4

Simulation Results and Analysis

4.1 Verification of MPLS Core Connectivity

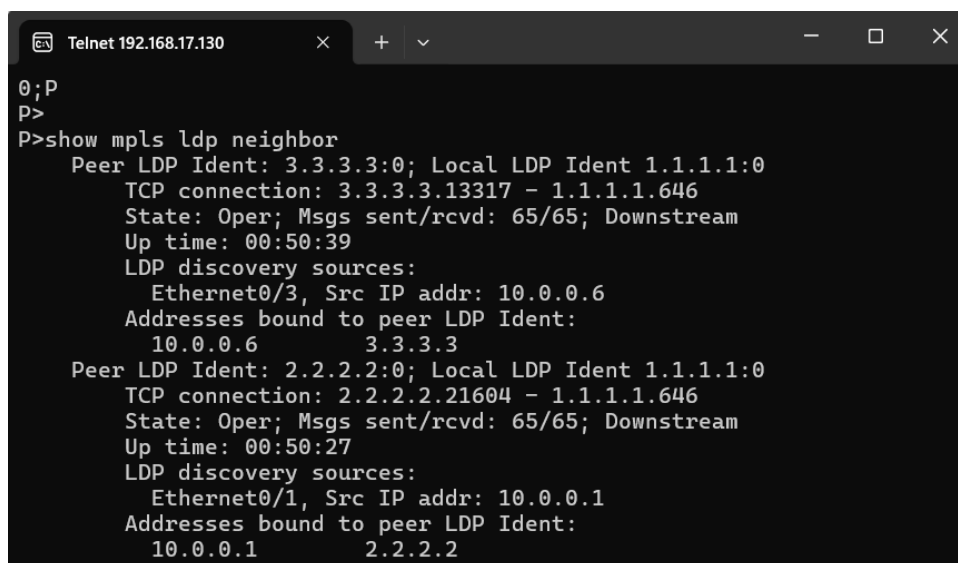
Before verifying VPN services, the MPLS backbone must be fully operational. The core routers have to exchange labels through LDP in order to establish valid Label Switched Paths (LSPs).

4.1.1 LDP Neighbor Status

The LDP session establishment is verified on the P router. An *Operational* status indicates that the P router successfully negotiated parameters and is exchanging label bindings with PE1 and PE2.

Command used on P Router

```
P# show mpls ldp neighbor
```

A screenshot of a Telnet window titled 'Telnet 192.168.17.130'. The window shows a command-line interface for a P router. The user enters '0;P' and 'P>'. Then, the command 'P>show mpls ldp neighbor' is entered. The output displays two LDP neighbor sessions. The first session is with Peer LDP Ident 3.3.3.3:0, Local LDP Ident 1.1.1.1:0, TCP connection 3.3.3.3.13317 - 1.1.1.1.646, State: Oper, Msgs sent/rcvd: 65/65, Downstream, Up time: 00:50:39. The second session is with Peer LDP Ident 2.2.2.2:0, Local LDP Ident 1.1.1.1:0, TCP connection 2.2.2.2.21604 - 1.1.1.1.646, State: Oper, Msgs sent/rcvd: 65/65, Downstream, Up time: 00:50:27. Both sessions show LDP discovery sources as Ethernet0/3 and Ethernet0/1 respectively, with IP addresses 10.0.0.6 and 10.0.0.1, and addresses bound to the peer LDP Ident as 10.0.0.6/3.3.3.3 and 10.0.0.1/2.2.2.2.

```
Telnet 192.168.17.130
0;P
P>
P>show mpls ldp neighbor
  Peer LDP Ident: 3.3.3.3:0; Local LDP Ident 1.1.1.1:0
    TCP connection: 3.3.3.3.13317 - 1.1.1.1.646
    State: Oper; Msgs sent/rcvd: 65/65; Downstream
    Up time: 00:50:39
    LDP discovery sources:
      Ethernet0/3, Src IP addr: 10.0.0.6
    Addresses bound to peer LDP Ident:
      10.0.0.6      3.3.3.3
  Peer LDP Ident: 2.2.2.2:0; Local LDP Ident 1.1.1.1:0
    TCP connection: 2.2.2.2.21604 - 1.1.1.1.646
    State: Oper; Msgs sent/rcvd: 65/65; Downstream
    Up time: 00:50:27
    LDP discovery sources:
      Ethernet0/1, Src IP addr: 10.0.0.1
    Addresses bound to peer LDP Ident:
      10.0.0.1      2.2.2.2
```

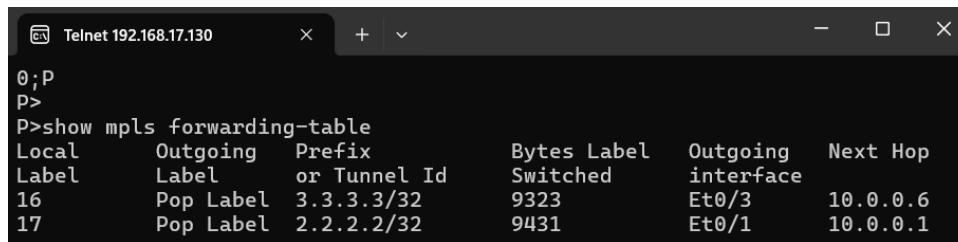
Figure 4.1: LDP Neighbor Status on P Router

4.1.2 MPLS Forwarding Table

To confirm that core routers are switching packets based on labels rather than conventional IP lookups, the MPLS forwarding table (LFIB) is examined.

Command used on P Router

```
P# show mpls forwarding-table
```



The screenshot shows a Telnet window titled 'Telnet 192.168.17.130'. The user enters '0;P' and 'P>'. Then, the command 'P>show mpls forwarding-table' is entered, resulting in the following table output:

Local Label	Outgoing Label	Prefix or Tunnel Id	Bytes Switched	Label	Outgoing interface	Next Hop
16	Pop Label	3.3.3.3/32	9323		Et0/3	10.0.0.6
17	Pop Label	2.2.2.2/32	9431		Et0/1	10.0.0.1

Figure 4.2: MPLS Forwarding Table on P Router

4.2 Verification of VPN Operations

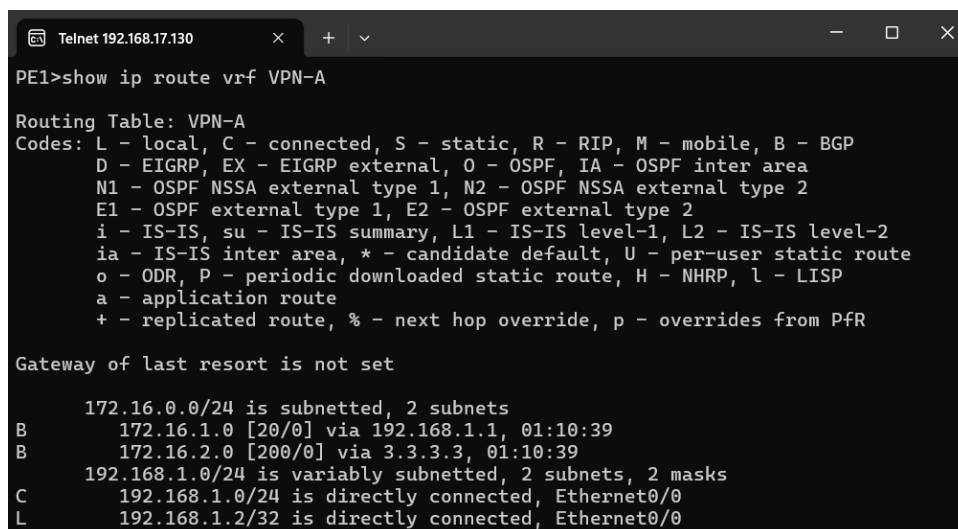
This section validates the control plane of the Layer 3 VPN by checking route isolation in VRFs and route distribution through MP-BGP.

4.2.1 VRF Routing Tables

The routing table of the customer instance VPNA is inspected on PE routers to confirm complete separation from the global routing table.

Command used on PE1

```
PE1# show ip route vrf VPNA
```



The screenshot shows a Telnet window titled 'Telnet 192.168.17.130'. The user enters 'PE1>show ip route vrf VPN-A'. The output shows the routing table for VRF VPN-A, including a legend for codes and a list of routes.

Routing Table: VPN-A
 Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

B	172.16.0.0/24 is subnetted, 2 subnets
	172.16.1.0 [20/0] via 192.168.1.1, 01:10:39
B	172.16.2.0 [200/0] via 3.3.3.3, 01:10:39
	192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C	192.168.1.0/24 is directly connected, Ethernet0/0
L	192.168.1.2/32 is directly connected, Ethernet0/0

Figure 4.3: VRF VPNA Routing Table on PE1

Analysis shows that PE1 learns the remote LAN prefix 172.16.2.0/24 through BGP. The route is received over the VPNv4 session from PE2 (3.3.3.3), confirming that the MP-BGP control plane operates correctly.

4.2.2 MP-BGP VPNv4 Routes

To further analyze route transport, the MP-BGP database is verified. This reveals VPNv4 prefixes together with their Route Distinguisher and Route Target information.

Command used on PE1

```
PE1# show bgp vpnv4 unicast all
```

```

0:PE1
PE1>
PE1>show bgp vpnv4 unicast all
BGP table version is 4, local router ID is 2.2.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
               x best-external, a additional-path, c RIB-compressed,
               t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
Route Distinguisher: 65001:1 (default for vrf VPN-A)
*>  172.16.1.0/24    192.168.1.1             0         0 65101 i
*>i  172.16.2.0/24    3.3.3.3                 0        100 0 65101 i

```

Figure 4.4: MP-BGP VPNv4 Table on PE1

4.3 End-to-End Connectivity Testing

The final and most critical verification step is testing data-plane connectivity between the two customer sites.

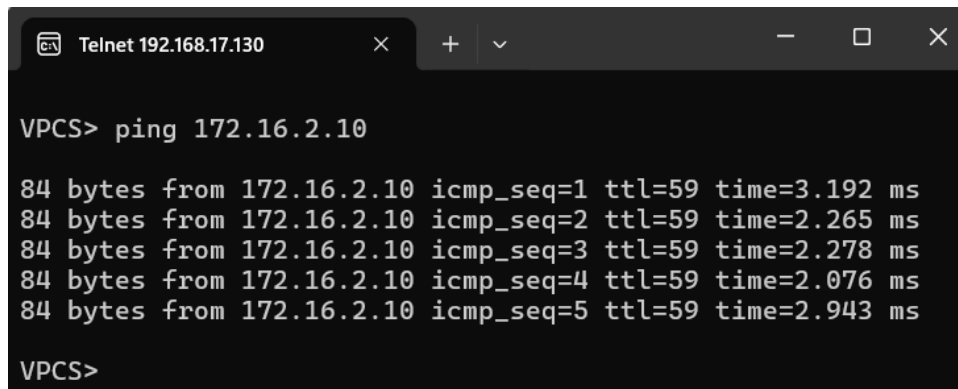
4.3.1 Ping Tests Between Customer Branches

The first test is performed from VPC 8 at Site 1 to VPC 10 at Site 2.

- **Source:** VPC 8 (172.16.1.10)
- **Destination:** VPC 10 (172.16.2.10)

Command used on VPC 8

```
VPCS> ping 172.16.2.10
```



```

Telnet 192.168.17.130
VPCS> ping 172.16.2.10

84 bytes from 172.16.2.10 icmp_seq=1 ttl=59 time=3.192 ms
84 bytes from 172.16.2.10 icmp_seq=2 ttl=59 time=2.265 ms
84 bytes from 172.16.2.10 icmp_seq=3 ttl=59 time=2.278 ms
84 bytes from 172.16.2.10 icmp_seq=4 ttl=59 time=2.076 ms
84 bytes from 172.16.2.10 icmp_seq=5 ttl=59 time=2.943 ms

VPCS>

```

Figure 4.5: Successful Ping from VPC 8 (Site 1) to VPC 10 (Site 2)

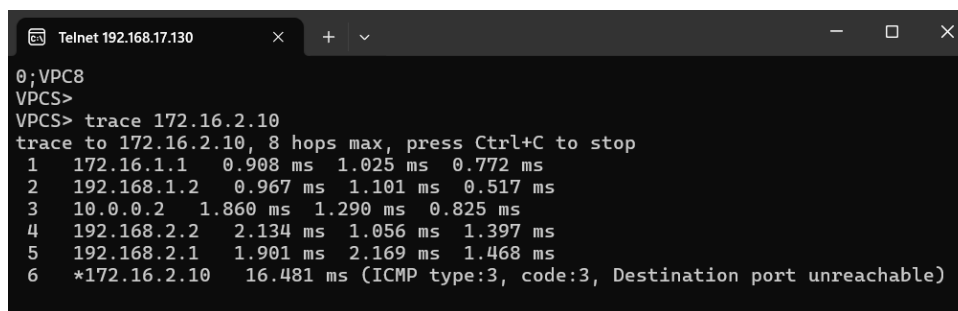
The successful ping response confirms full reachability between the two remote LANs. Request packets traverse Site 1, the service-provider MPLS core, and Site 2; reply packets return successfully to the source host.

4.3.2 Traceroute Analysis

To verify that traffic traverses the MPLS backbone through the expected forwarding path, a traceroute is performed from VPC 8 to VPC 10.

Command used on VPC 8

```
VPCS> trace 172.16.2.10
```



```

Telnet 192.168.17.130
0;VPC8
VPCS>
VPCS> trace 172.16.2.10
trace to 172.16.2.10, 8 hops max, press Ctrl+C to stop
 1  172.16.1.1    0.908 ms  1.025 ms  0.772 ms
 2  192.168.1.2   0.967 ms  1.101 ms  0.517 ms
 3  10.0.0.2     1.860 ms  1.290 ms  0.825 ms
 4  192.168.2.2   2.134 ms  1.056 ms  1.397 ms
 5  192.168.2.1   1.901 ms  2.169 ms  1.468 ms
 6  *172.16.2.10 16.481 ms (ICMP type:3, code:3, Destination port unreachable)

```

Figure 4.6: Traceroute from Site 1 to Site 2 Verifying MPLS Path

The traceroute output illustrates the packet flow across the network:

- **Hop 1 and 2:** the packet leaves Customer Site 1 and reaches PE1 through standard IP routing.
- **Hop 3:** the P router (10.0.0.2) shows MPLS label usage, confirming label switching inside the backbone.
- **Hop 4 and 5:** the packet reaches PE2, the label is removed, and the packet is forwarded to CE2 and then to VPC 10.

4.4 Troubleshooting and Challenges

During implementation and verification, several technical issues were encountered. The major problems, probable causes, and corrective actions are summarized below.

4.4.1 Configuration Persistence on PNETLab

Symptom: the message `% BGP not active` appeared, or device configuration was lost after a system reboot.

Cause: the running configuration was not saved to the startup configuration, and the configuration file was not exported to persistent PNETLab storage.

Solution: the `write` command was executed on the device CLI and the `Export CFG` function was used on the corresponding PNETLab node.

4.4.2 BGP AS-Path Loop Prevention

Symptom: ping tests failed with `Request timeout` despite an established BGP neighbor state.

Cause: both customer sites use the same Autonomous System number (AS 65101). As a result, CE2 detects its own AS in the received AS-path and discards the route advertisement to prevent a loop.

Solution: the `neighbor <IP> as-override` command is configured on PE routers so the provider AS replaces the customer AS in the advertised path before forwarding updates to the remote CE router.

CHAPTER 5

Conclusion and Future Works

5.1 Conclusion

This project successfully designed and simulated a comprehensive MPLS Layer 3 VPN network using the PNETLab emulation platform. Through the implementation of OSPF, LDP, and MP-BGP, a robust service-provider backbone was established to transport customer data securely and efficiently.

The results in Chapter 4 provide clear evidence of network functionality:

- **Core connectivity:** LDP neighbor relationships were established successfully, and the forwarding table confirmed label-based packet forwarding.
- **Service isolation:** VRF and MP-BGP provided logical customer separation, and routing-table verification confirmed the isolation of customer routes from the global table.
- **End-to-end reachability:** ping and traceroute tests verified successful communication between customer sites and demonstrated MPLS label usage across the backbone.

Overall, the project demonstrates that MPLS VPN is a scalable and reliable solution for modern enterprise networks. It addresses the limitations of traditional peer-to-peer VPNs by simplifying customer routing while enabling a service provider to support multiple isolated customers on shared infrastructure.

5.2 Future Works

Although the project successfully implemented a functional MPLS VPN, the topic offers several promising directions for further study:

- **Quality of Service (QoS):** configure traffic classification and policing to prioritize latency-sensitive applications such as voice and video.
- **MPLS Traffic Engineering (MPLS-TE):** replace shortest-path forwarding with explicit path steering, for example via RSVP-TE, to optimize bandwidth utilization.

- **IPv6 over MPLS (6VPE):** extend the current design to support IPv6 VPN services.
- **Integration with SD-WAN:** study how MPLS backbones can interoperate with SD-WAN overlays to create hybrid WAN architectures.

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